

--	--	--	--	--	--	--	--

***B. Tech. Degree VI Semester Special Supplementary Examination
January 2016***

ME 1605 CAD/CAM
(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Differentiate wire frame from solid modelling.
(b) What is Detroit type of automation?
(c) What are the advantages of CNC?
(d) Describe a rotary displacement transducer, with a neat sketch.
(e) Write a note on static and dynamic errors in CNC tools.
(f) What is CNC tooling.
(g) Write a note on artificial intelligence.
(h) What are the control approaches of robots?

PART B

(4×15 = 60)

- II. (a) Explain the data exchange between CAD and CAM. (8)
(b) Describe design database. (7)
- OR**
- III. (a) Explain the analysis of automated flow lines. (8)
(b) Describe high volume discrete parts production. (7)
- IV. (a) Explain the control theory of CNC tools. (7)
(b) What are the different sensors and transducers used in CNC machines? (8)
- OR**
- V. What is CAPP? Explain structure and use of major CAPP languages. (15)
- VI. Explain the special design features of CNC tools. (15)
- OR**
- VII. Describe ATC and APC. (15)
- VIII. (a) Explain the various sensors used in robots. (7)
(b) Describe drives and control systems used in robots. (8)
- OR**
- IX. (a) Describe in detail, FMS. (7)
(b) Write a note on robotic programming languages. (8)